



FLORIDA STATE UNIVERSITY
DEPARTMENT OF SCIENTIFIC COMPUTING

WRITTEN PRELIMINARY EXAMINATION
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Question - 1

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1 Fast Fourier Transformation[Dr. Meyer-Baese]

1.1 Part 1

n_1	n_2			
	0	1	2	3
0	x[0]	x[3]	x[6]	x[9]
1	x[4]	x[7]	x[10]	x[1]
2	x[8]	x[11]	x[2]	x[5]

k_1	k_2			
	0	1	2	3
0	X[0]	X[9]	X[6]	X[3]
1	X[4]	X[1]	X[10]	X[7]
2	X[8]	X[5]	X[2]	X[11]

Figure 1: Tables for index map.

1.2 Part 2

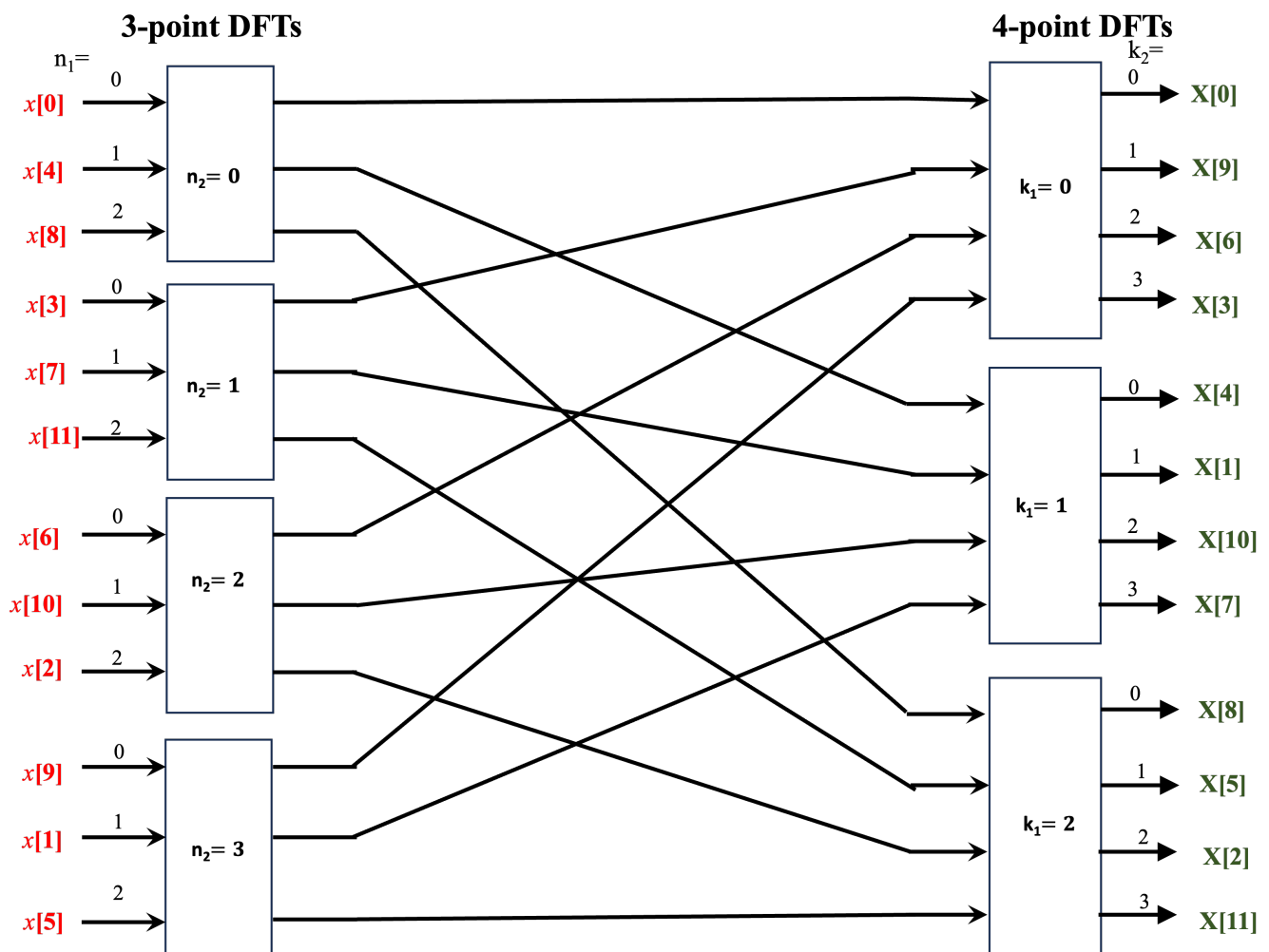


Figure 2: SFG for given FFT.

For completing this question, I have taken help from the examples give in [1].

References

- [1] Uwe Meyer-Baese and U Meyer-Baese. *Digital signal processing with field programmable gate arrays*, volume 65. Springer, 2007.